PC 4: Dog Messenger on a Local Network

TCP Sockets, Concurrency, QR Codes, and AI Store

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Core Architecture and TCP Concurrency

- **Strict LAN Environment:** The system automatically filters out virtual network adapters (Docker, VMware) to guarantee connection strictly over the physical interface.
- **Non-Blocking TCP Sockets:** The main communication flows through port **8189** without freezing the graphical interface.
- **Thread Concurrency:** Every connected client is assigned a dedicated `Thread`. This ensures the server can broadcast messages and files simultaneously without data collisions.

Custom Protocol and Smart Cloning

- **Payload Formatting:** We designed a protocol delimited by | and URL-encoded. It handles `MSG` (chat), `FILE` (Base64 file transfer), and `STORECFG` (background configuration).
- **QR Linking:** Instant network access by scanning `dogmsg://IP:PORT`.
- **Silent Mode (`?clone=1`):** A Smart Cloning parameter that transfers the active session to a new device seamlessly, inheriting credentials without triggering a connection broadcast.

System Scalability: The Virtual Store Node

- **Independent Service Node:** To demonstrate horizontal scaling, we deployed a completely separate server (`Server50Tienda`) operating on port **8190**.
- **Controlled Disconnection:** Accessing the store forces the client to save its state, cleanly destroy the chat socket, and open an exclusive TCP tunnel for commercial queries.
- **State Restoration:** Exiting the store automatically reconnects the user to the main chat, restoring their previous state seamlessly.

Bayesian AI and State Machine

Automated Sales Assistant

Built to process natural language without relying on third-party cloud APIs.

- **Probabilistic Classifier:** A Naive Bayes model programmed in pure Python deduces user intent mathematically from the socket text.
- **Hybrid Validation:** The AI handles open questions, but a **Finite State Machine (FSM)** takes control during checkouts to enforce strict binary responses.
- **Data Integrity:** Inventory transactions are protected by Mutex Locks, preventing concurrency issues during simultaneous purchases.

Technical Conclusions

- Developing TCP Sockets from scratch confirms the viability of building resilient distributed systems without high-level abstraction frameworks.
- Isolating services across different TCP ports proves the architecture is highly modular and ready for horizontal scaling.
- **Handover:** Next, my partner will explain the core architecture flow and perform a live demonstration of the codebase in action.

System Flow and Scalability Topology